IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MATHEMATICS 1B

Tuesday, 30 April 2:30 pm

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions. This exam has four questions, worth 25 Marks each. Do all questions.

NO CALCULATORS ALLOWED.

Examiners responsible First Marker(s) : D. Nucinkis
Second Marker(s) : W.T. Pike

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THE QUESTIONS

1. a) A line L through the origin has direction vector $\underline{\mathbf{b}} = (1,1,1)^T$. Find the projection of the vector $\underline{\mathbf{a}} = (1,2,3)^T$ onto/along the vector $\underline{\mathbf{b}}$. Use this to find a point Q on L, nearest to the point $P(1,2,3)$ and hence the distance from P to L. Provide a sketch and explain your reasoning! [5]

b) Show that both $u(x,y) = \sin(x^2 - y^2)$ and $v(x,y) = \cos(y^2 - x^2)$ are possible solutions for w in the partial differential equation (PDE)

$$y\frac{\partial w}{\partial x} + x\frac{\partial w}{\partial y} \;=\; 0\,.$$

Hence show that the function $z(x,y) \;=\; x^2u - y^2v$ satisfies the PDE

$$y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} \;=\; 2xy(u-v)\,.$$

[7]

c) A function of two variables is given as

$$f(x,y) \;=\; x^2\cos y\,.$$

(i) Obtain the Taylor series expansion of $f(x,y)$ about the point $(x_0,y_0) = (1,\pi)$, up to quadratic terms.

(ii) If we use this expansion to estimate $(1.1)^2\cos(3)$, consider the discarded terms of order three and higher, to give an upper bound for the size of the expected error. Explain your reasoning! [6]

d) Show that the first-order Ordinary Differential Equation

$$\frac{2x}{y} - \cos x \;=\; \left(\frac{x^2}{y^2} - 2e^{2y}\right)\frac{dy}{dx}$$

is exact and find a solution satisfying $y(0) = -1$. [7]

2. a) Evaluate the double integral

$$I = \int\int_R xy\, e^{x^2+y^2}\, dx\, dy\,,$$

(i) when R is a square with vertices at the origin, $(1,0),(0,1)$, and $(1,1)$; [4]

(ii) when R is in the first quadrant, bounded by the coordinate axes, and the unit circle $x^2+y^2=1$, changing to polar coordinates (r,θ) . You may assume the result $dxdy = rdrd\theta$ [5]

b) Let a function be defined as $f(x) = -x-1$ on the interval $[-1,0]$.

(i) Sketch $Eve(x)$ and $Odd(x)$, the even and odd periodic extensions of f, both with period $T=2$, for $-3 \le x \le 3$. [2]

(ii) If Fourier Series for $Eve(x)$ and $Odd(x)$ are calculated, what is the value each Fourier Series takes at $x=2$? State the rate of decay of the Fourier coefficients for each series. [3]

(iii) Obtain the Fourier Series for $Eve(x)$. [6]

(iv) Use (iii) and Parseval's Theorem to show that $\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$ [5]

3. a) Show (prove) that the Fourier transform of a cosine function is

$$\mathscr{F}[\cos(at)] = \pi[\delta(\omega - a) + \delta(\omega + a)]$$

where $\delta(x)$ is the Dirac delta-function. Derive a similar expression for the Fourier transform of $\sin(at)$. [7]

b) Use the definition of the Laplace transform to prove the first shift theorem, that if $\mathscr{L}[f(t)] = \bar{f}(s)$, then

$$\mathscr{L}[e^{at} f(t)] = \bar{f}(s-a) \, . \qquad [4]$$

(ii) Use partial fractions and the first shift theorem to obtain the following inverse Laplace transform:

$$\mathscr{L}^{-1}\left[\frac{10}{(s-2)(s^2+2s+2)}\right] \, . \qquad [7]$$

(iii) Use Laplace transforms and (ii), to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2} - \frac{dx}{dt} - 2x = 10e^{-t}\cos t$$

which satisfies the initial conditions $x(0) = 1$, and $x'(0) = 3$. [7]

4. a) A system of linear equations is given as

$$\begin{array}{rcrcrcl} -x & + & y & + & 2z & = & 0\,, \\ x & + & \alpha y & - & z & = & -3\,, \\ 2x & + & y & - & 6z & = & \beta\,, \end{array}$$

where $\alpha, \beta \in \mathbb{R}$ are parameters.

(i) Write the system in matrix form $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$, find the determinant of the matrix A, and give the value of α for which A is singular. [3]

(ii) Let α be the value found in (i). Use Gaussian elimination to reduce the augmented coefficient matrix $(A : \underline{\mathbf{b}})$ to echelon form, and find $N(A)$, the nullspace of A. What kind of geometrical object does the nullspace represent? [5]

(iii) Find the value of β for which the system of equations is consistent, explaining why this is necessary. Let β take this value. Now find a particular solution of the system, and hence the general solution, giving a geometric interpretation of the solution. [5]

b) A matrix is given as

$$A = \begin{pmatrix} 1 & -1 & 1 & -1 \\ 2 & -3 & 4 & 2 \\ -1 & 3 & -5 & -7 \end{pmatrix}.$$

Obtain the echelon form of A and state its rank. Hence find a basis for each of the four fundamental subspaces of A: nullspace, column space, row space, and left nullspace. In each case, give your reasoning! [12]